



Technical Service Bulletin: TSB230096

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ECM Reset and Communication Issue Troubleshooting

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QST30 CM552

Issue Summary

Symptom:

- ECM **not** able to communicate upon engine key on

Root Cause:

- ECM issue

Verification

Use the following steps to diagnose ECM:

1. With Calterm® (preferred) or INSITE™ electronic service tool and a datalink adapter hooked up to a suspect ECM.
2. Set battery voltage at 26V to 28V.
3. Key on and verify if the ECM will communicate. Key off. Repeat a few times.
4. If the ECM will **not** communicate key off and reduce battery voltage to 12V to 14V.
5. Key on and see if the ECM will communicate.
6. If the ECM will **not** communicate at the higher voltage but does at the lower voltage, the ECM likely has the reset issue.
7. A malfunctioning ECM has been detected when keyed on at a higher voltage and **not** communicating will be continuously resetting.

Note : If using Calterm® and can communicate, check the reset parameter NRESETSE and see how many resets have occurred. A normally functioning module should **not** have any or **only** a few resets. Improper shutdowns (removing battery voltage in less than 30 sec after keying off) will increment the reset count each occurrence. The

longer the ECM is left on and not communicating the higher the reset count will be when the ECM does communicate at the lower voltage. The reset count can increment by hundreds in just a few minutes.

Fault codes that can be seen for suspect ECMs are listed below:

Table 1, Fault Codes	
Fault Code	Description
285	SAE J1939 Multiplexing PGN Timeout Error - Abnormal Update Rate
346	Engine Control Module Warning Software error - Bad intelligent device or component
426	SAE J1939 Datalink - Data Erratic, Intermittent, or Incorrect; J1939 Network #1 - Data erratic, intermittent or incorrect
434	Power Lost With Ignition On - Data Erratic, Intermittent, or Incorrect
1117	Power Lost With Ignition On - Data Erratic, Intermittent, or Incorrect
1135	SAE J1939 Data Link 2 Engine Network - Abnormal Update Rate
2727	SAE J1939 Data Link 2 Engine Network - Abnormal Update Rate

Resolution

Replace malfunctioning ECM. See QST30 Industrial Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666214. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/87/87-019-031.html)

Document History

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2023-5-3	Module Created

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